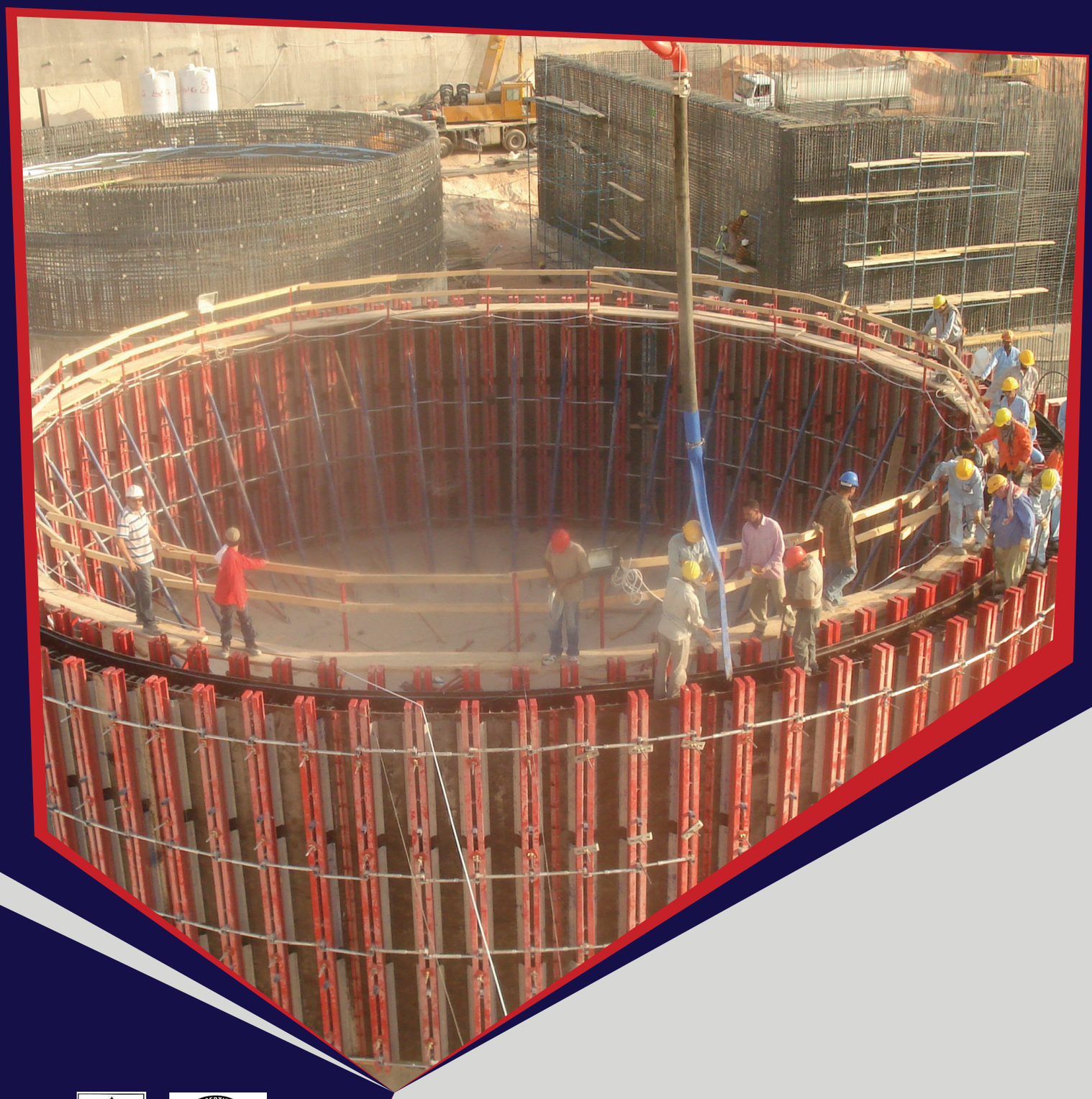


UKAS QUALITY MANAGEMENT 005

SYSTEM CERTIFICATION ISO 9001:2008 SGS

ACROW

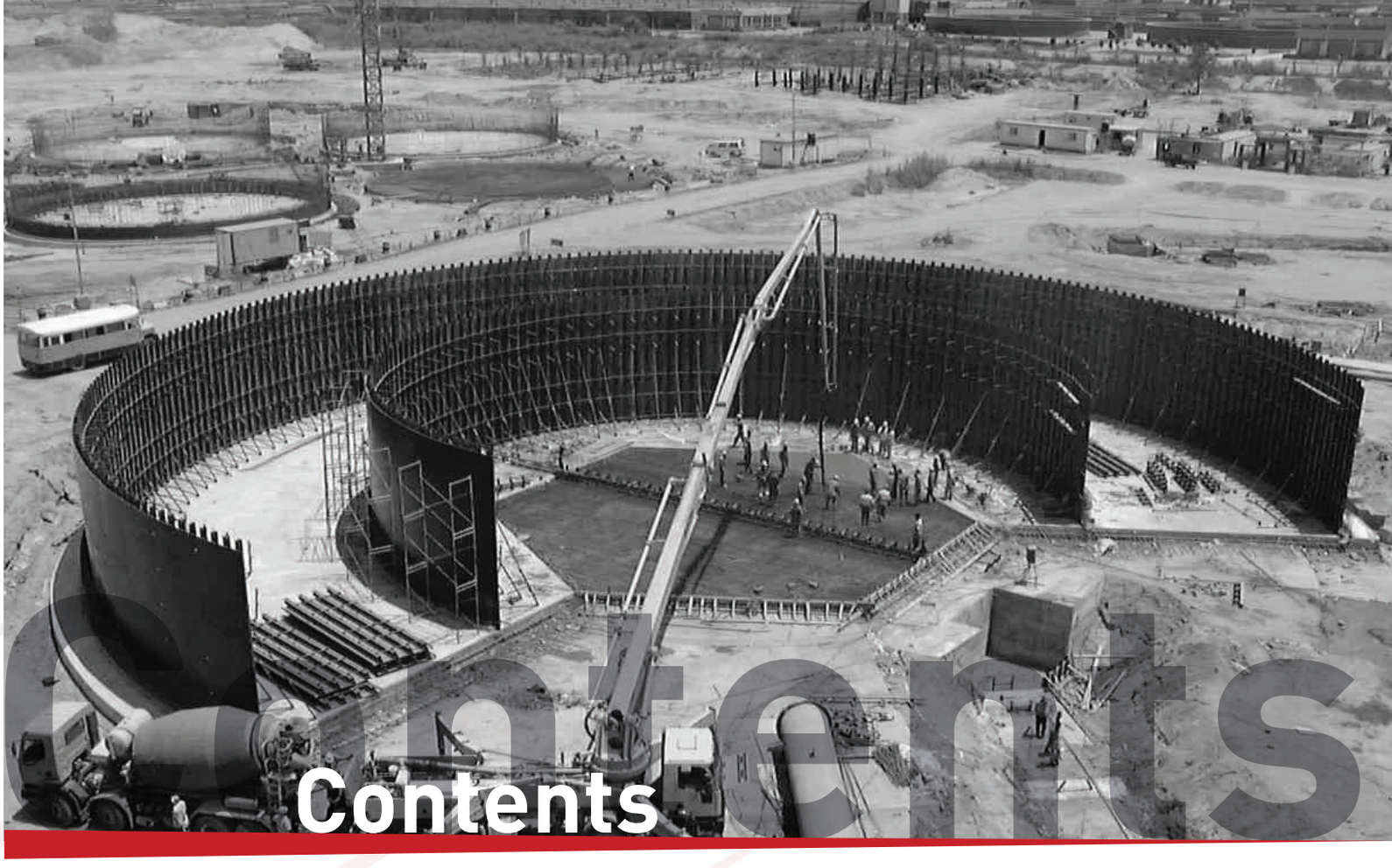
Formwork Technology

Vari Form System

ACROW MISR Provides Full Site Service Based On More Than 35 Years Experience In The Field



ACROW
Formwork Technology



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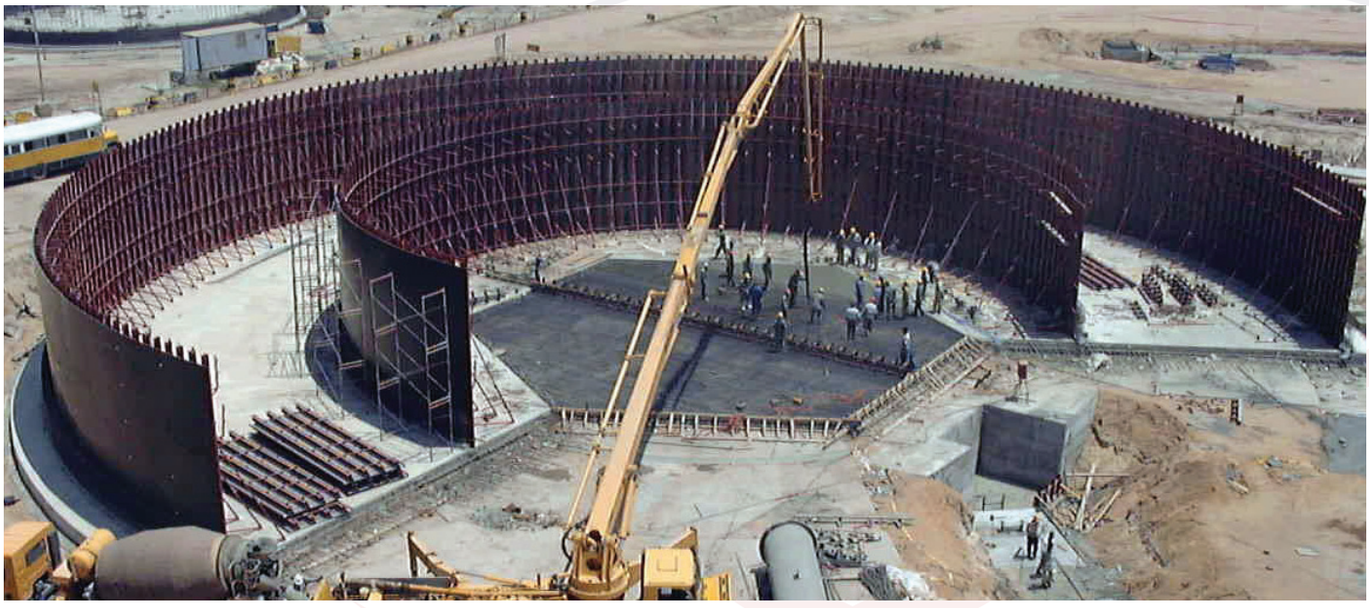
ACROW Formwork Technology

Introduction

Being one of the largest formwork and scaffolding companies in Egypt, We are introducing Vari Form System. It has proved to be very flexible especially as a curved wall formwork system. In addition, it could be easily adjusted by turn-buckle to allow for the formation of complex curves and shapes. Finally, it is designed for curves till a diameter of minimum 3.00 m.

Advantages:

- Provides a superior concrete finish.
- Needs only two turnbuckles every 3.6 m height.
- No special purpose walings.
- Allows for any wall height.
- Needs simple standard components.
- It could be used as single sided shutter.





Application

Vari form system can be used horizontally for closed circle or as a part of a curve. It can be also used vertically for one shot or several stages on platform. Vari form is suitable for many applications including:

- Curved Walls
- Water Treatment Tanks
- Large Diameter Columns
- Bullnosed piers
- Circular shafts
- Silos





Applications







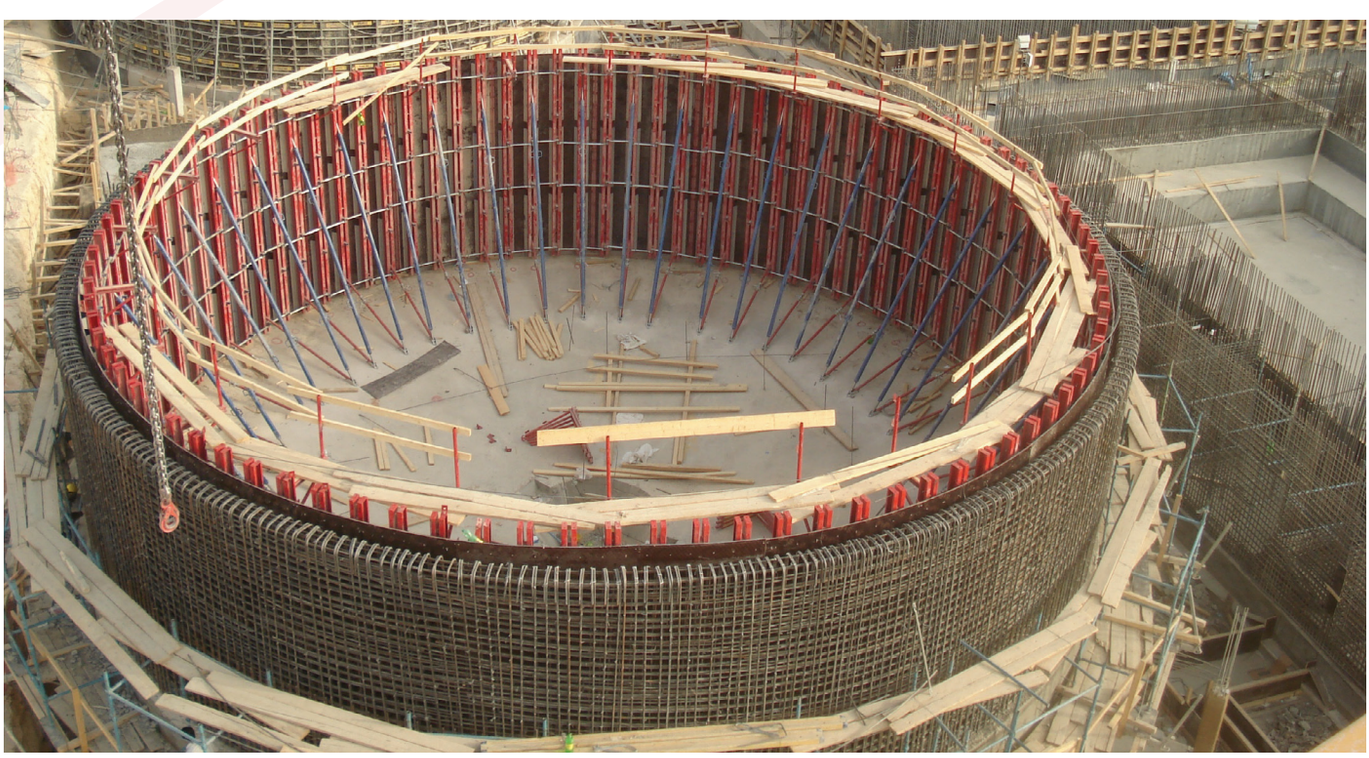
Loadings

•It is recommended to be used for concrete pressures up to 60 KN/m², where 18mm fin ply or similar is used.

•DW 16 tie load capacity is 90 KN.
•Acrow open web solider mechanical properties are as per attached table.

•Concrete pressure can be calculated using acrow concrete pressure cataloge

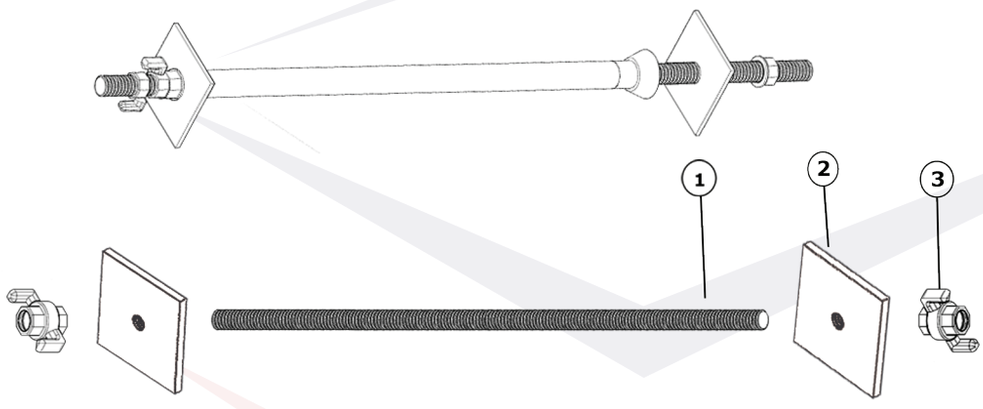




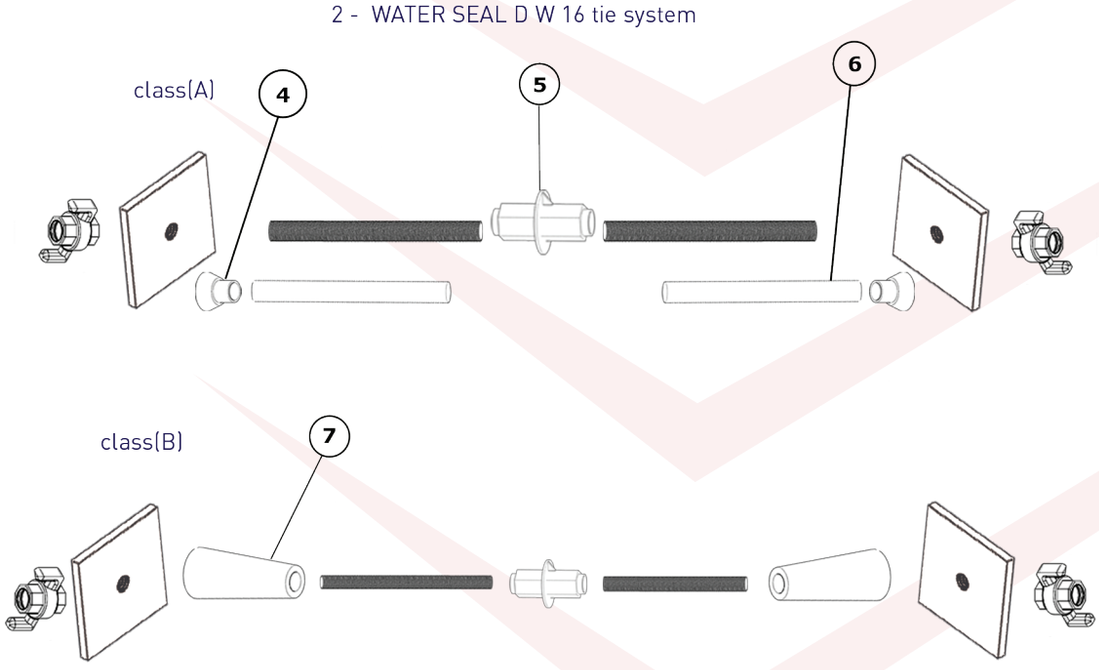
Dw 16 Tie System

•The acrow DW 16-ties system is primarily used with soldiers, where minimum number of ties is required.
•The continuous thread high tensile steel rod is the basis of the DW 16 tie system assembly.
The wing nut is designed to freely spin on the fast threaded rod, which effectively speeds both the erection and stripping times of the formwork.

1- THRU DW tie system: there are no lost items.



2 - WATER SEAL D W 16 tie system



1- D W16 tie rod 2- Steel washer 3- Wing nut 4- Plastic cone 5-Water barrier 6- Palstic tube 7- Steel cone

Components



Vari form trough section:
•The length of trough section is 900 mm and it has a 300 mm increment.
Vari form turn-buckles:
•Only two turn-buckles are used per solider length up to 3.60 m.
Vari form edge angle:
•The panels are joined together by using bolt M 16 * 50 or M 16 * 100.
Open web solider:
•The table below shows the technical information and the loading data of the open web soldier.

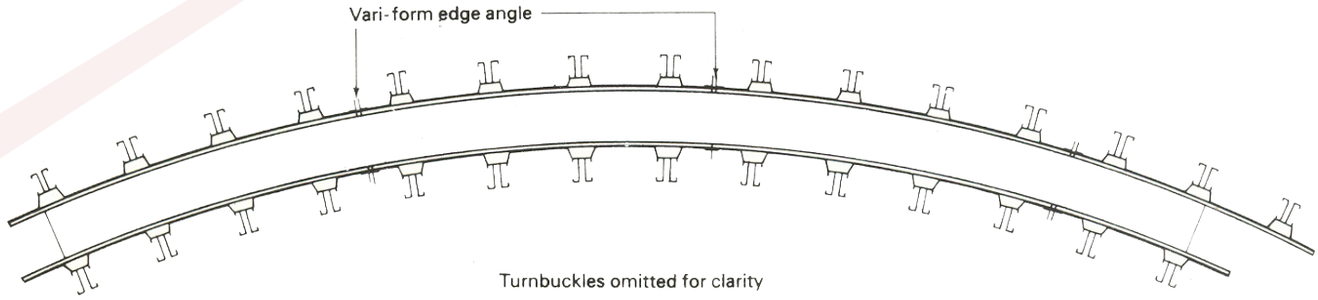
Open Web Soldier

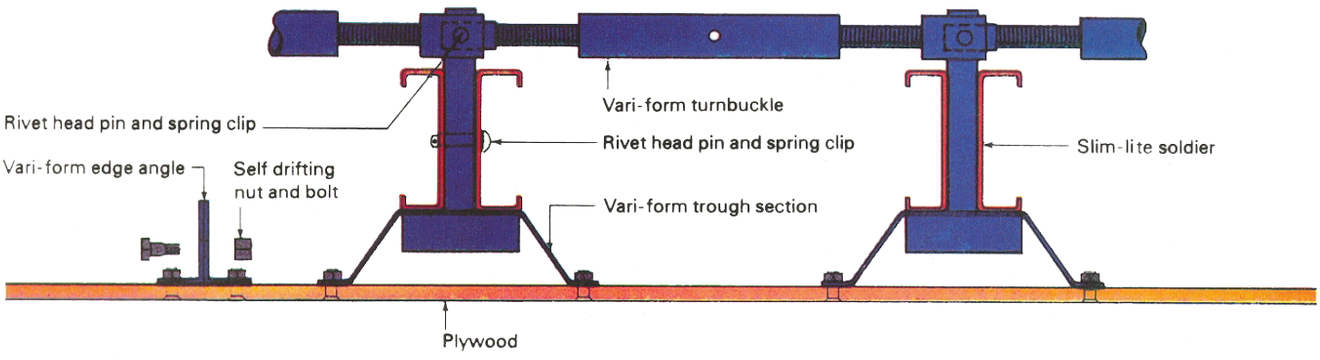
OPEN WEB SOLDIER (HxBxbxt)	A cm^2	wt kg/m'	X-X			Y-Y		
			I_x cm^4	Z_x cm^3	i_x cm	I_y cm^4	Z_y cm^3	i_y cm
160x70x30x3	20.42	15.70	777.70	97.12	6.23	566.00	62.19	5.32
160x80x80x4	29.62	23.25	1143.60	143.00	6.21	1115.60	110.45	6.13

Curved Forms

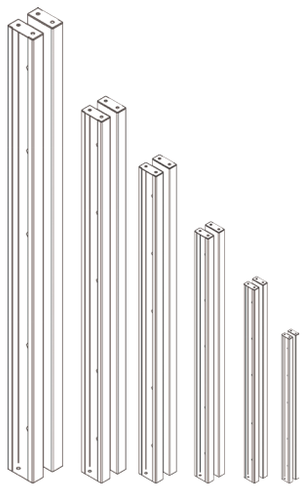
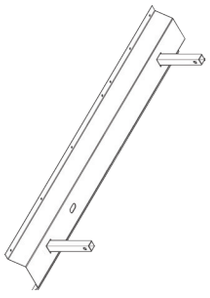
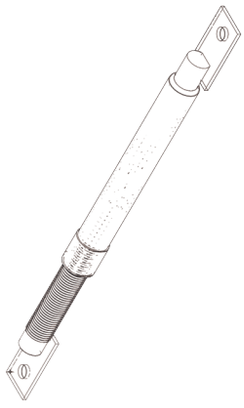
Application details

Curved form

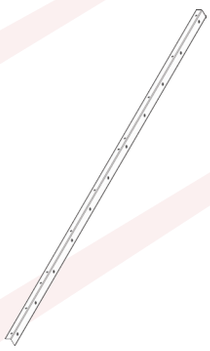
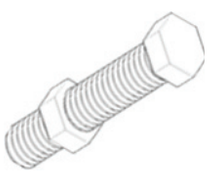




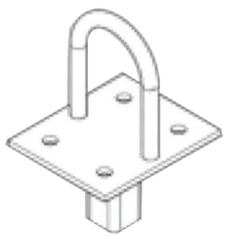
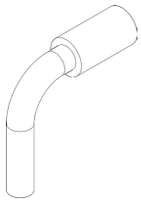
Components

Item	Description	Code no.	Weight (kg)
	Open web Soldier C16, L=0.90m	6221177011895	37.38
	Open web Soldier C16, L=1.20m	6221177011901	49.85
	Open web Soldier C16, L=1.50m	6221177011918	62.31
	Open web Soldier C16, L=1.80m	6221177011925	74.77
	Open web Soldier C16, L=2.10m	6221177011932	87.23
	Open web Soldier C16, L=2.40m	6221177011949	99.70
	Open web Soldier C16, L=2.70m	6221177011956	112.16
	Open web Soldier C16, L=3.00m	6221177011963	124.62
	Open web Soldier C16, L=3.30m	6221177011970	137.08
	Open web Soldier C16, L=3.60m	6221177011987	149.55
	Open web Soldier C16, L=3.90m	6221177011994	162.01
	Open web Soldier C16, L=4.20m	6221177012007	174.47
	Open web Soldier C16, L=4.50m	6221177012014	186.93
	Open web Soldier C16, L=4.80m	6221177012021	200.00
	Open web Soldier C16, L=5.10m	6221177012038	211.86
	Open web Soldier C16, L=5.40m	6221177012045	224.32
	Open web Soldier C16, L=5.70m	6221177012052	236.78
	Open web Soldier C16, L=6.00m	6221177012069	249.25
	Trough Section, L=0.90 m	6221177024468	10.875
	Trough Section, L=1.20 m	6221177024475	14.50
	Trough Section, L=1.50 m	6221177024482	17.40
	Trough Section, L=1.80 m	6221177024499	20.35
	Trough Section, L=2.10 m	6221177024505	24.70
	Trough Section, L=2.40 m	6221177024512	27.60
	Trough Section, L=2.70 m	6221177024429	32.625
	Turn - buckle (50-70) cm	6221177014346	4.50

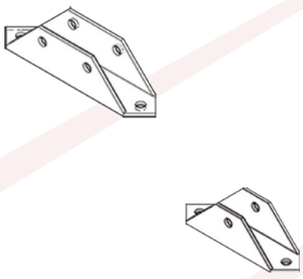
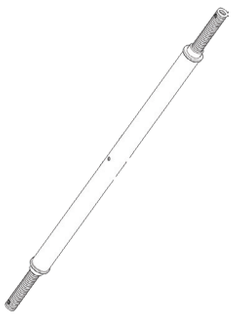
Components

Item	Description	Code no.	Weight (kg)
	Rivet Pin 16 mm, L =7.50 cm Rivet Pin 17 mm, L =12.5 cm Rivet Pin 17 mm, L =11.5 cm	6221177012281 6221177012304 6221177012298	0.12 0.20 0.18
	Spring clip large Spring clip small	6221177022617 6221177022624	0.03 0.01
	Edge Angle, L =0.90 m Edge Angle, L =1.20 m Edge Angle, L =1.50 m Edge Angle, L =1.80 m Edge Angle, L =2.10 m Edge Angle, L =2.40 m Edge Angle, L =2.70 m Edge Angle, L =3.00 m Edge Angle, L =3.30 m Edge Angle, L =3.60 m Edge Angle, L =3.90 m	6221177009038 6221177009045 6221177009052 6221177009069 6221177009076 6221177009083 6221177009090 6221177009106 6221177009113 6221177009120 6221177009137	3.95 5.27 6.58 7.90 9.22 10.54 11.85 13.17 14.50 15.81 17.12
	Bolt m 16 X100 mm Bolt m 16 X50 mm	2012090028565 2012090028572	0.184 0.130

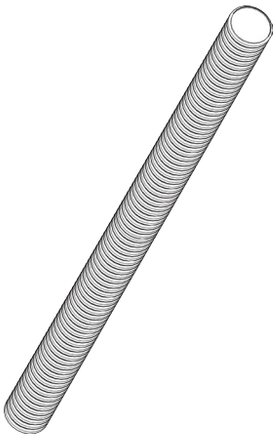
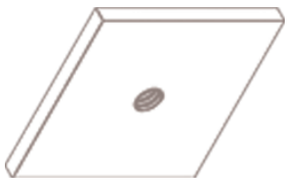
Components

Item	Description	Code no.	Weight (kg)
	Lifting Eye For C16	6221177011123	1.50
	Hi Skaf Braket L = 2.00 m L = 1.20 m	 6221177003289 6221177003265	 46.00 39.00
	Hi Skaf Anchor	6221177028084	2.85

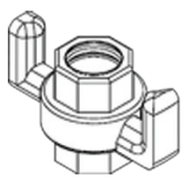
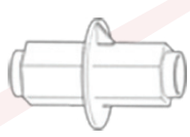
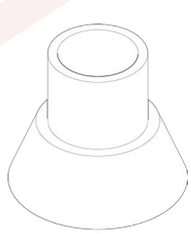
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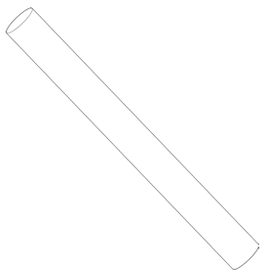
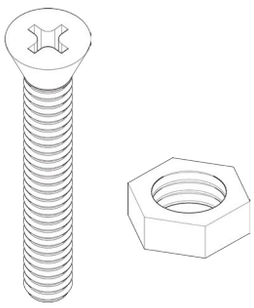
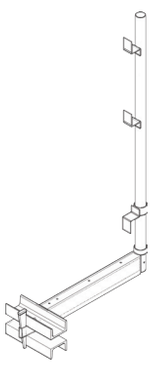
Item	Description	Code no.	Weight (kg)
	Scaffolding Tube,L=1.00 m Scaffolding Tube,1.5"X2.9mm L = 2.00 Scaffolding Tube,1.5"X2.9mm L = 3.00 Scaffolding Tube,L=1.50 m Scaffolding Tube,L=2.00 m Scaffolding Tube,L=2.50 m	6221177017354 6221177017361 6221177017378 6221177017385 6221177017392 6221177017408	3.35 6.70 10.05 4.395 5.86 7.32
	Double Base Plate Single Base Plate	6221177000813 6221177000882	2.4 1.45
	Turn Buckle Push Pull PPS 131 Push Pull PPS 132 Push Pull PPS 161 Push Pull PPS 162 Push Pull PPS 201 Push Pull PPS 202 Push Pull PPS 252 Push Pull PPS 302 Push Pull PPS 352	6221177014254 6221177014261 6221177014278 6221177014285 6221177014292 6221177014308 6221177014315 6221177014322 6221177014339	9.00 11.40 9.83 12.43 13.14 14.00 15.70 19.00 23.10

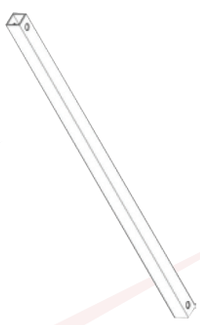
Components

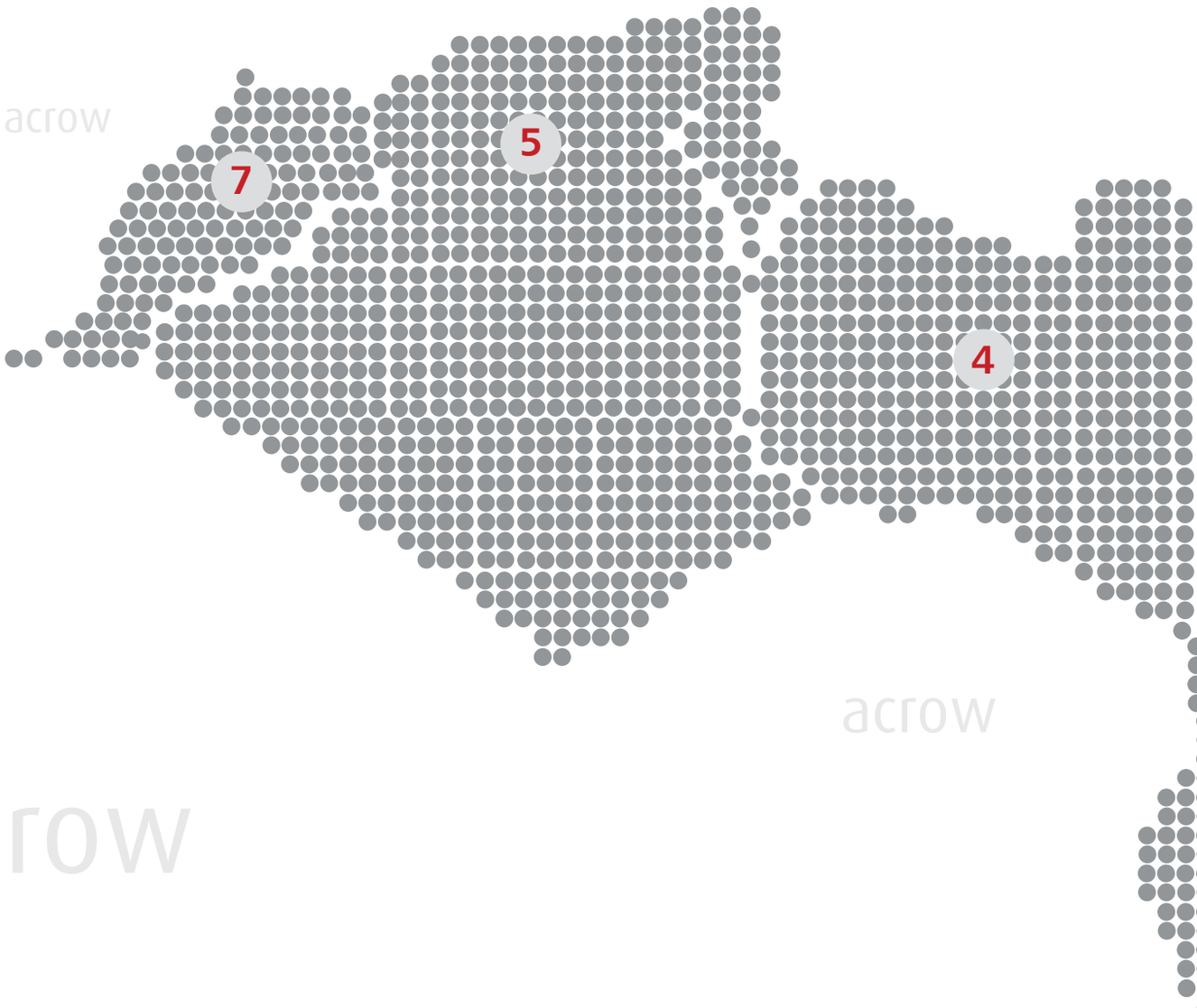
Item	Description	Code no.	Weight (kg)
	Tilt up shore x0 Tilt up shore x1 Tilt up shore x2 Tilt up shore x3 Tilt up shore x4	6221177023942 6221177023959 6221177023966 6221177023973 6221177023980	10.90 17.50 18.65 21.25 25.64
	DW Tie 16mm, L = 1.00 m DW Tie 16mm, L = 1.50 m DW Tie 16mm, L = 1.75 m DW Tie 16mm, L = 2.00 m DW Tie 16mm, L = 3.00 m DW Tie 16mm, L = 4.50 m DW Tie 16mm, L = 6.00 m	6221177023140 6221177023157 6221177023164 6221177023171 6221177023188 6221177023195 6221177023201	1.58 2.37 2.77 3.16 4.74 7.11 9.48
	Steel Washer	6221177027513	1.1

Components

Item	Description	Code no.	Weight (kg)
	Wing Nut DVD	6221177011383	0.45
	Water Barrier (flange type) (type 1)	6221177028220	0.52
	Plastic Cone 12.5 mm	2014100032746	0.003

Item	Description	Code no.	Weight (kg)
	Plastic Tube PVC, L = 1.0 m	6221177028152	0.32
	Counter Sunk Bolt	2012080028407	0.077
	Access Bracket (for Vertical Solider)	6221177003135	10.88

Item	Description	Code no.	Weight (kg)
	Diagonal Tube 40X40X2.5 mm, L =1.0 m	6221177008864	2.82
	Shore Connecter (For Soldier C16)	6221177006365	1.12



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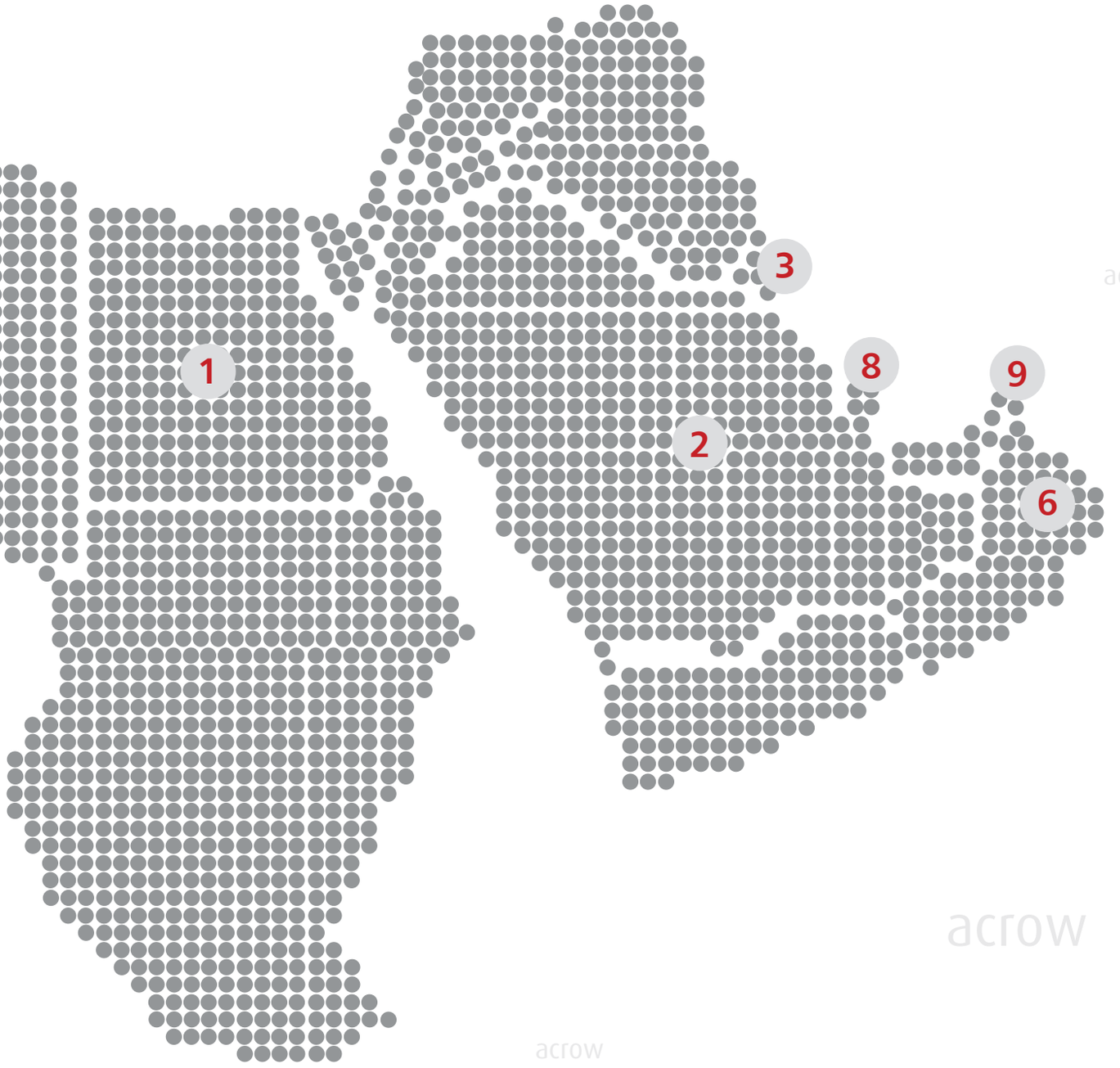
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ACROW
Formwork Technology


Acrow H20 & solider


Acrow One shutter solider


Acrow V.M.C


Acrow Tech-Form


Acrow Cuplock


Acrow Bridge table form

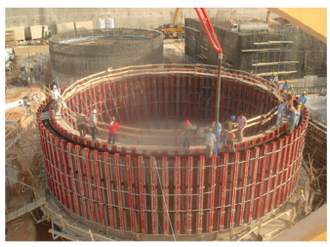

Acrow Circular column

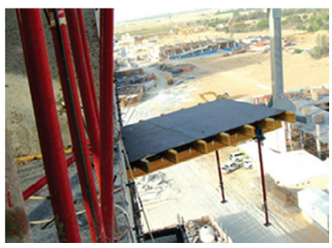

Acrow climbing system


Acrow shorbrace Table form


Acrow Big steel panel


Acrow European prop


Acrow Vari form


Acrow Prop table form


Acrow frame


Acrow early striking system


Acrow Ring lock

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02/2013, Code No.: 2007050060068